PART 02 · CHAPTER 08

Classical chart patterns

Translate familiar silhouettes into definitions that can fail.

Answer edition: includes continuation, worked solutions and the knowledge-check answer.

beginner About 7 min of prose · practice takes additional time AI editorial review 2026-10-12 · v0.4

▼ Review scope

Re-accepted for documented whitespace or shared study locator/date correction; earlier substantive review retained.

Recorded by /root. The review covers editorial evidence; it does not establish learning by human readers.

> THE IDEA TO KEEP
>
> Pattern names organize historical geometry. Completion criteria, timing, and a fair comparison matter more than resemblance.

By the end of this chapter

- Translate a familiar shape into candidate, recognition and completion states.
- Apply peak tolerance, separation, and support-break rules to supplied closes.
- Distinguish pattern completion from a measured target outcome.
- Recognize the main geometric families and reasonable alternative readings.

Builds on: Trends, swings & changing structure · Support, resistance & price areas

Study a small core before the catalogue

A classical chart pattern names an arrangement of several price swings. This chapter's core is a range, converging boundaries, and a complete double-top worked analysis. The optional catalogue then shows flags, pennants, mirrored bottoms, and head-and-shoulders geometry so the reference remains useful without making memorization a prerequisite for chapter 9. A clean description states which observations count and when a candidate becomes complete.

A cutoff note records the state using only observations available at that time. In this worked case, stop after the first eleven completed closes and keep that note before opening the later branches. The same prefix must receive the same initial classification in every branch; changing that initial record after reading the ending would substitute hindsight for analysis.

The double-top rule sheet

For this closing-price example, select peaks and troughs using strict two-before/two-after neighbors, as in chapter 5. Require two confirmed peaks separated by at least four session steps. Their absolute price difference must be at most 1% of the first peak. The minimum close between them is the intervening reference; our example also waits for its local-trough confirmation. Completion requires a subsequent completed close strictly below that reference. A later higher close above the first peak before completion invalidates this candidate version.

These tolerances and delay choices are the chapter's teaching rules. They make "roughly equal tops" reproducible, rather than claiming a single canonical detector. A discretionary reader might select wick highs or a wider tolerance. That alternative needs its own record. Our clean example lets you see every comparison before trying one where the answer depends on a borderline choice.

The full supplied decision rule

Choice	Declared version
Input	Completed daily closes
Swing selection	Strict two earlier/two later closes
Peak separation	At least 4 session steps
Peak tolerance	Difference ≤1% of first peak
Intervening reference	Minimum intervening close; wait for trough confirmation
Completion	Later completed close below that reference
Candidate invalidation before completion	Close above first peak

First pass: classify the D11 record before D12 exists

The rule sheet above supplies the detector; the table and chart below contain all eleven completed daily closes available at the D11 cutoff. No D12–D15 values, branch outcomes or derived swing labels are supplied. Use the unchanged rule, not a more convenient tolerance chosen after reading the continuation.

Identify every peak and trough that has enough earlier and later neighbors by D11. For the eligible peak pair, record its session spacing, percentage price difference using the first peak as denominator, intervening reference and each recognition time. Then classify the D11 state: no qualifying pair, candidate, completed or invalidated.

Write the numerical conditions that a later completed close would have to satisfy for completion or pre-completion invalidation. Say how equality at each boundary is treated. Distinguish the numerical observation from an executable short trade; no borrowing, order, quote, cost or fill model is supplied.

Freeze that note, then compare it with “Work from the prices to the candidate” after revealing the continuation. Read each later branch separately from the same D11 state. The branches are deliberately constructed contrasts, not frequencies from which to estimate success probabilities.

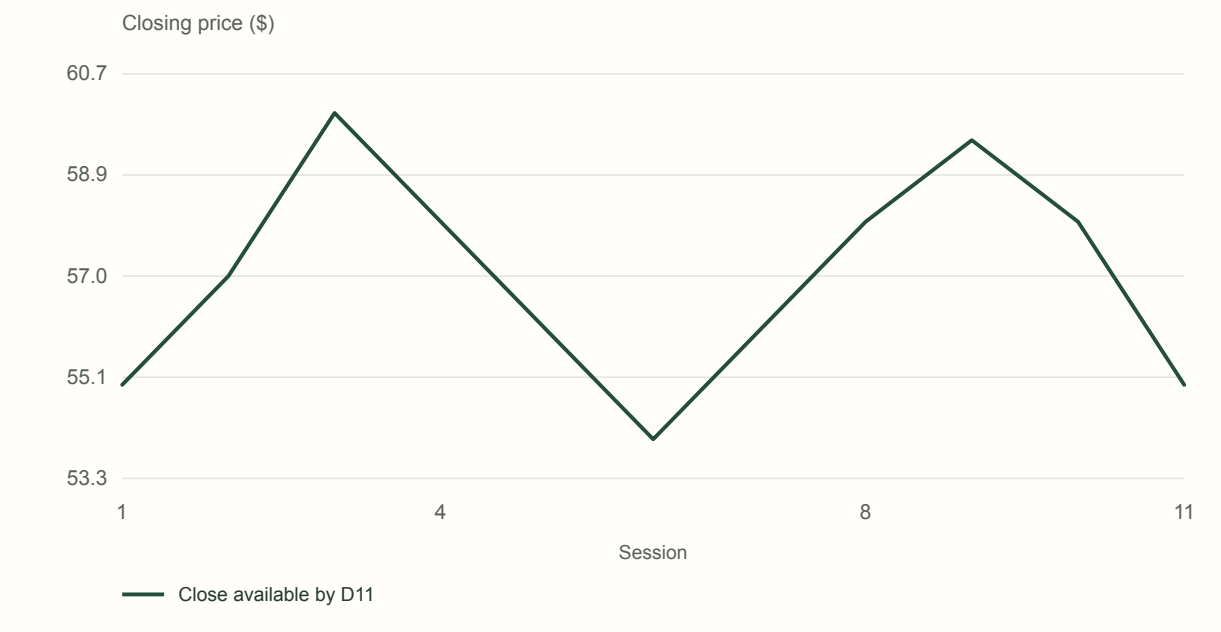
Completed closes available after D11 (fictional USD)

Session	Close ($)
D1	55
D2	57
D3	60
D4	58
D5	56
D6	54

Session	Close ($)
D7	56
D8	58
D9	59.5
D10	58
D11	55

FICTIONAL TEACHING ILLUSTRATION

The unlabelled D1–D11 close path



Fictional completed closes only. The reader must construct the swings and any transition boundary; no D12 continuation, peak annotation or answer line is included.

▼ View the illustration's data

The unlabelled D1–D11 close path

Session	Close available by D11
1	55
2	57
3	60
4	58

Session	Close available by D11
5	56
6	54
7	56
8	58
9	59.5
10	58
11	55

Compare the continuation with your frozen note. This reference includes outcome headings, complete ledgers and solutions. In the learner print edition it starts on a new page after the cutoff packet.

Work from the prices to the candidate

D3 reaches $60 and D5 confirms it as a peak. D6 reaches $54 and D8 confirms it as a trough. D9 reaches $59.50; D10$58 and D11$55 confirm that peak after D11. The peak separation is 9 −3 = 6 steps, exceeding four. Their difference is $0.50; relative to 60 it is 0.50/60 ≈0.833%, below 1%. The intervening minimum is 54, already recognized before the second peak.

At D11 the candidate is valid under the stated construction, but its completed-close transition has not occurred. The latest close 55 remains above 54. A partially finished analysis should say “candidate double top; completion boundary 54; recognition cutoff D11.” It should not say that the support break has occurred merely because the two highs look similar.

Why exclude other observations? D2$57 is not a peak because D3 is higher; D4$58 is not a trough because D5 and D6 are lower. D8$58 is not a separate peak because D9 is higher. The declared comparisons choose a small set from the supplied path and prevent arbitrary anchor selection.

Chronological construction

Observation	Value ($)	Rule comparison	Known after
D3 first peak	60	Above two earlier/later closes	D5
D6 trough	54	Below two earlier/later closes	D8
D9 second peak	59.5	Above two earlier/later closes	D11
Peak spacing	6 steps	9 −3 ≥4	D11
Peak difference	0.833%	0.50/60 × 100 ≤1%	D11
D11 close	55	Not below 54	D11: candidate only

Shared prefix: freeze both branches after D11

Session	Close ($)
D1	55
D2	57

Session	Close ($)
D3	60
D4	58
D5	56
D6	54
D7	56
D8	58
D9	59.5
D10	58
D11	55

Reveal completion, failure and an unfulfilled projection

In branch A, D12 closes 53.50, below 54, and completes the declared pattern. In branch B, D12 closes 61.50, above the first peak 60, invalidating the candidate before any completed close below 54. Those are different states from the same initial information. The branch B rise does not show that the original arithmetic was wrong; it shows why the construction and transition must be recorded separately.

A conventional double-top measured reference subtracts the first peak-to-trough height from the boundary. Height 60 −54 = $6; reference 54 −6 = $48. Now give the proposed outcome a clock: “Does a close at or below 48 occur within the next three completed sessions after the D12 break?” Branch A’s next closes 52, 54, 55 do not meet it. The pattern completed, while this particular target-and-horizon condition did not.

A reader using a longer horizon could reach a different outcome on additional data. A reader asking whether an intraday low reached 48 would need different fields. Our three-session closing outcome is a separate teaching rule, not part of the geometric formula. Neither classification supplies an executable short-sale plan or its costs. Chapter 9 develops execution with a long breakout case; advanced chapters evaluate broader strategies.

Classify different questions, rather than one win/lose label

Question	Branch A	Branch B
Candidate known after D11?	Yes	Yes
D12 completed close below 54?	Yes:53.50	No:61.50

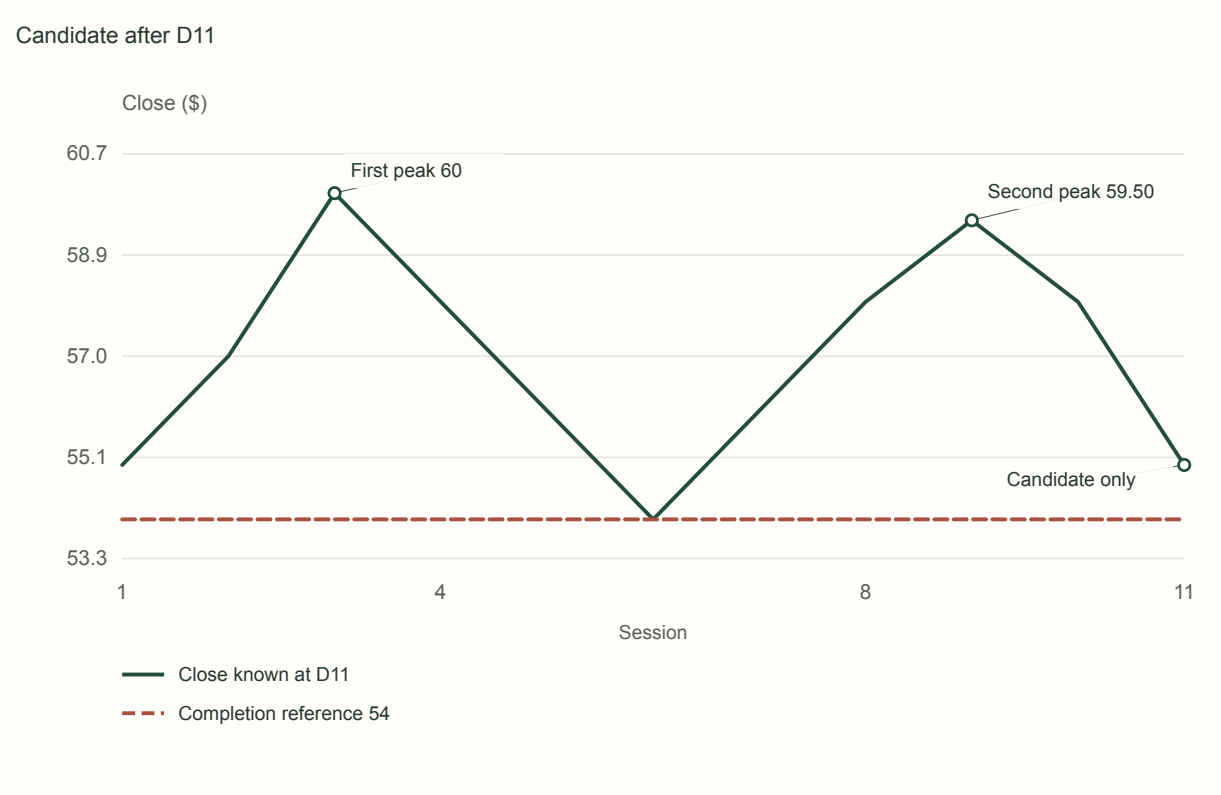
Question	Branch A	Branch B
Close above 60 before completion?	No	Yes:61.50
Next 3 closes ≤48 after completion?	No:52, 54, 55	No completed-pattern event to score

Alternative reveals after the identical prefix

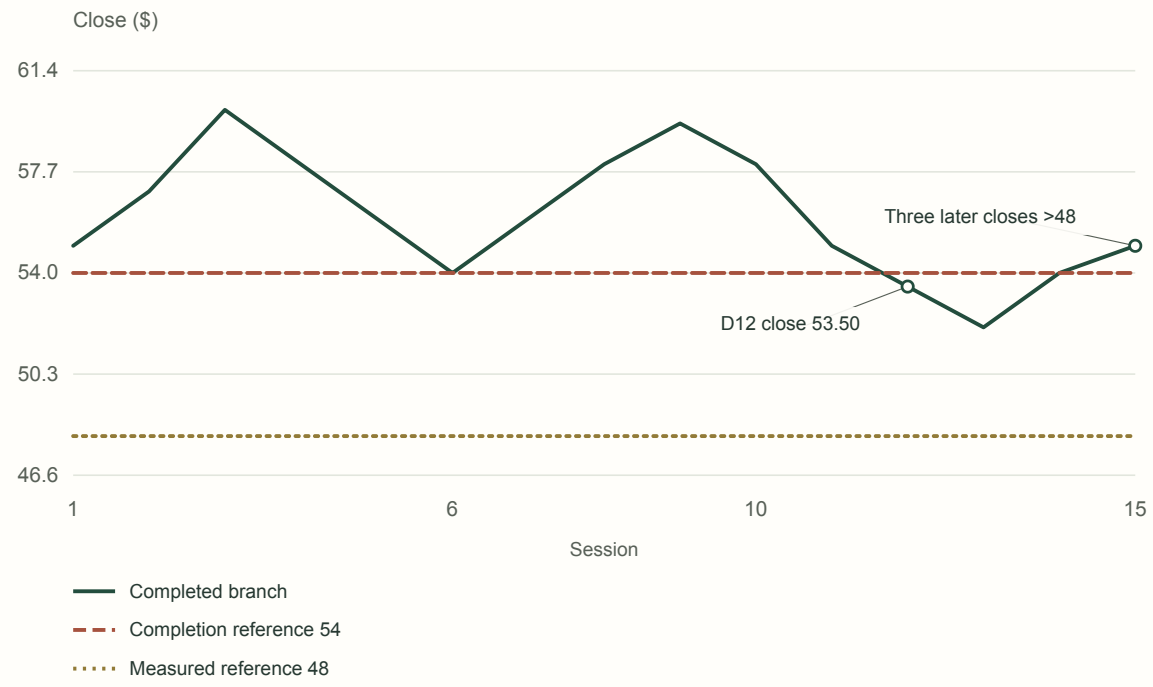
Session	Branch A close ($)	Branch B close ($)
D12	53.5	61.5
D13	52	62
D14	54	61
D15	55	62

FICTIONAL TEACHING ILLUSTRATION

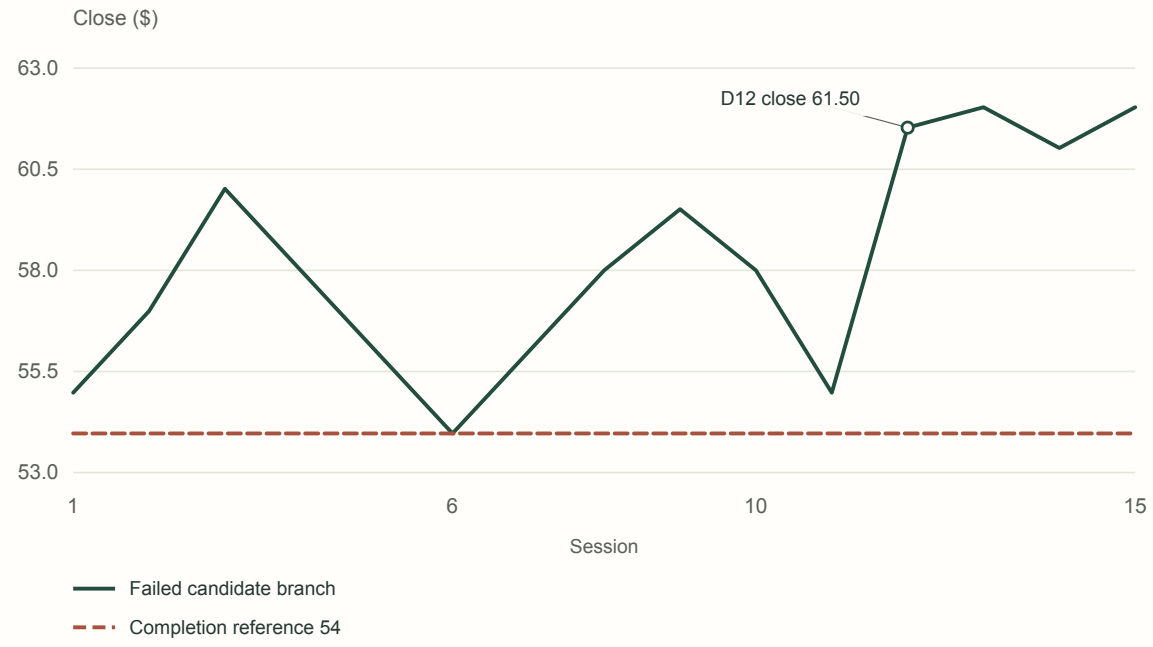
Identical candidate, two different later branches



Branch A: completed on D12, reference not reached



Branch B: new high invalidates the candidate



Supplied closes share D1–D11. Peaks 60 and 59.50 fit the declared tolerance; trough 54 is the completion boundary. The candidate at D11, completed branch A, and invalidated branch B are distinct states.

▼ View the illustration's data

Candidate after D11

Session	Close known at D11	Completion reference 54
1	55	54
2	57	54
3	60	54
4	58	54
5	56	54
6	54	54
7	56	54
8	58	54
9	59.5	54
10	58	54
11	55	54

Branch A: completed on D12, reference not reached

Session	Completed branch	Completion reference 54	Measured reference 48
1	55	54	48
2	57	54	48
3	60	54	48
4	58	54	48
5	56	54	48
6	54	54	48
7	56	54	48
8	58	54	48
9	59.5	54	48
10	58	54	48
11	55	54	48
12	53.5	54	48
13	52	54	48

Session	Completed branch	Completion reference 54	Measured reference 48
14	54	54	48
15	55	54	48

Branch B: new high invalidates the candidate

Session	Failed candidate branch	Completion reference 54
1	55	54
2	57	54
3	60	54
4	58	54
5	56	54
6	54	54
7	56	54
8	58	54
9	59.5	54
10	58	54
11	55	54
12	61.5	54
13	62	54
14	61	54
15	62	54

Ranges and triangles use the same boundary discipline

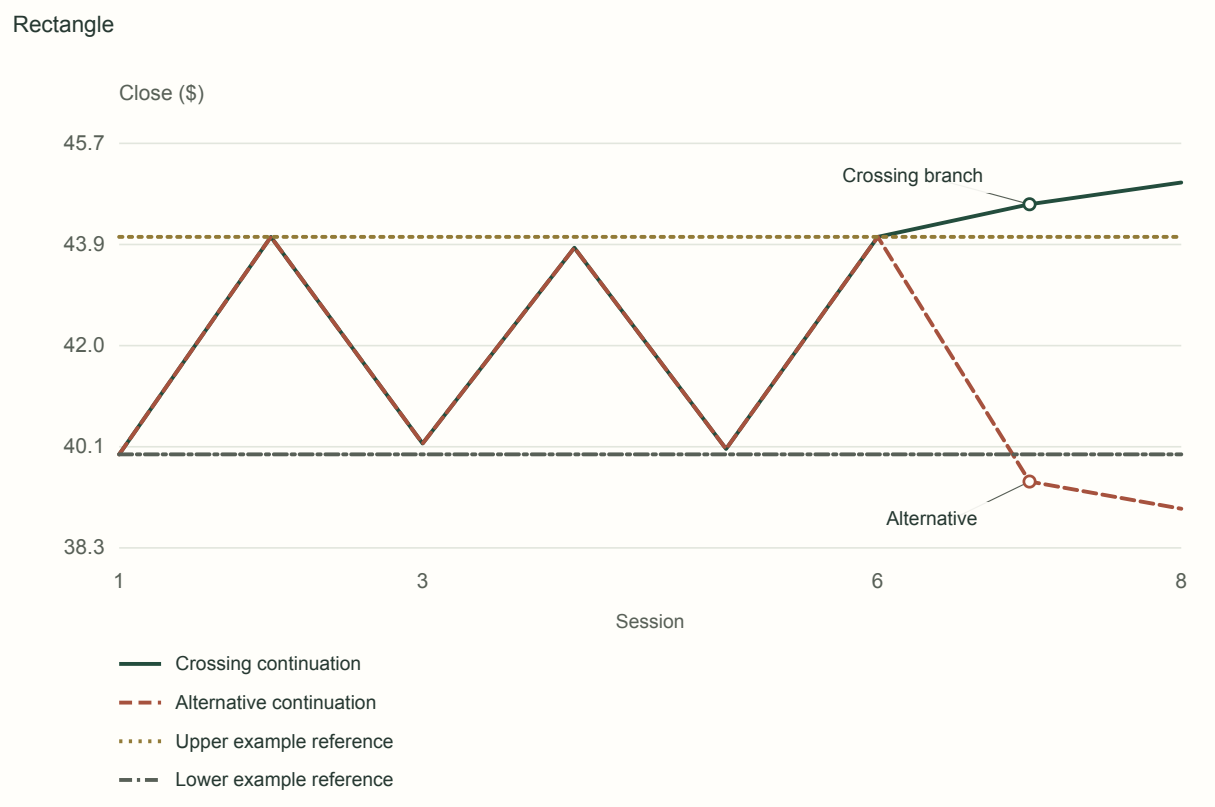
A rectangle organizes repeated reactions between approximately horizontal upper and lower areas. For the supplied rectangle panel, closes 40, 44, 40.20, 43.80, 40.10, 44 establish a lower area near 40 and upper area near 44. At the sixth close, freeze a teaching upper boundary 44 and lower boundary 40. A later close 44.60 completes an upward closing-cross condition; a competing later 39.50 close exits downward instead. The six-price prefix alone chooses neither direction.

For the symmetrical-triangle panel, selected highs 50, 49, 48 pair with lows 44, 45, 46. Their separation narrows 6 →4 →2. Fix the last upper/lower reference at 48/46 for this simplified completion exercise rather than silently extending a discretionary sloping line. A next 48.50 close exits above the last upper reference; 45.50 exits below the last lower reference. A true sloping-boundary detector would need dated anchors and projected values, as chapter 5 demonstrated.

Ascending and descending triangles make one boundary approximately horizontal and the other sloping. Their small panels display the selected sequences. Do not infer direction from the category name alone: define the boundary and observe the actual crossing. Contraction is a description of the range; the next transition is another observation. The guided and independent tasks ask you to apply those ideas to new values.

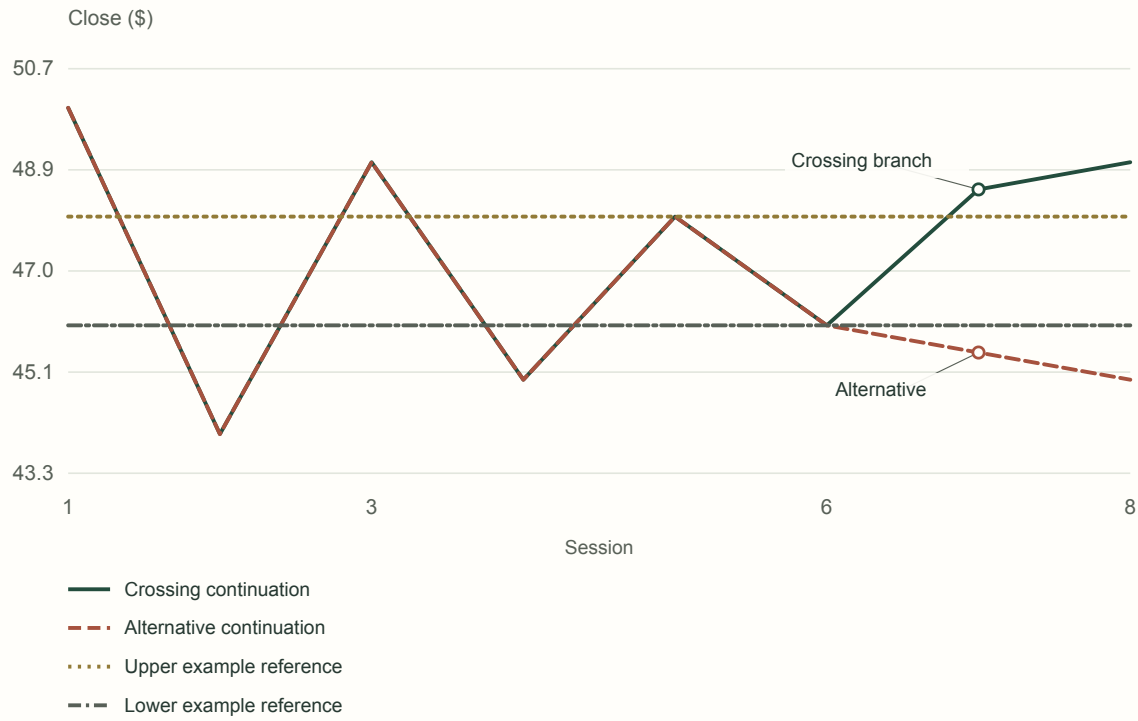
FICTIONAL TEACHING ILLUSTRATION

Boundary families: candidate prefix and alternative exits



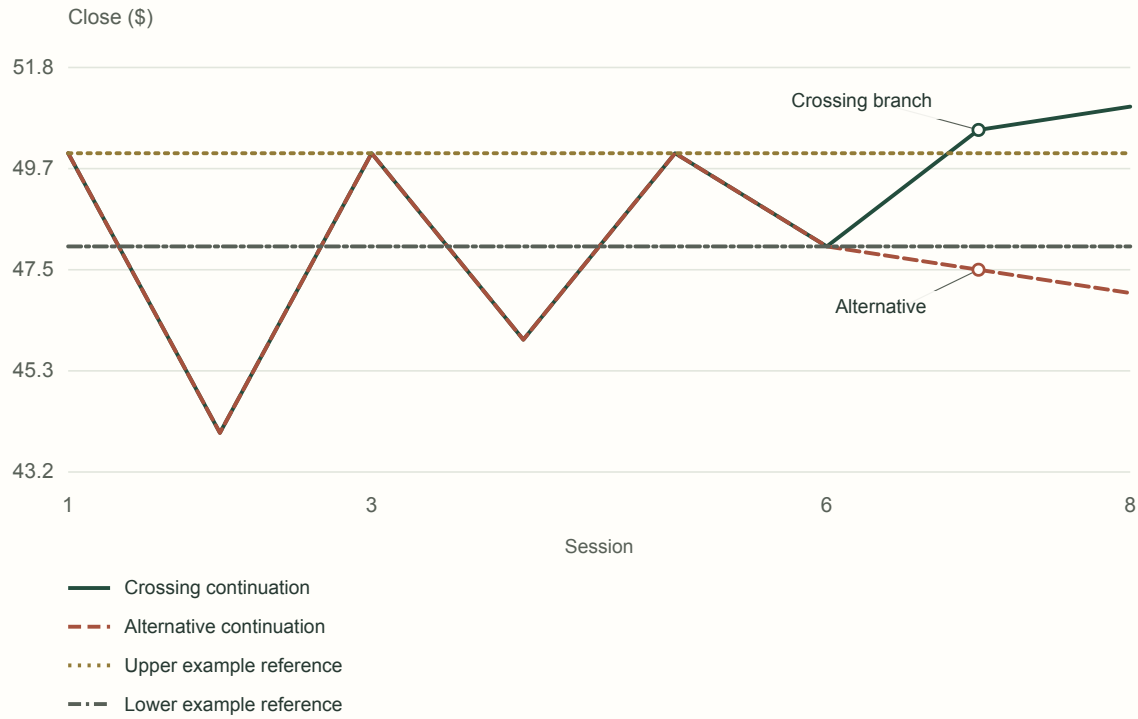
Same six-point range precedes the alternative closes.

Symmetrical triangle



Selected ranges narrow; these teaching crossings use the final 48/46 reference values.

Ascending triangle



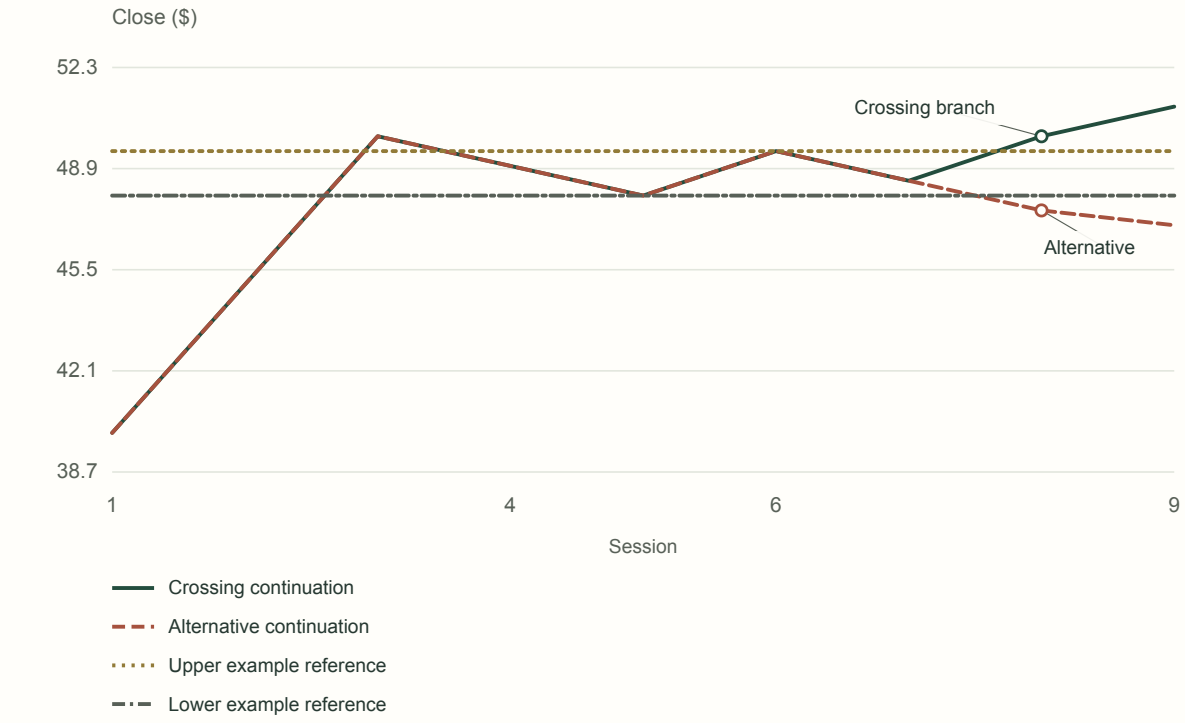
Horizontal selected highs 50 and rising troughs 44, 46, 48.

Descending triangle

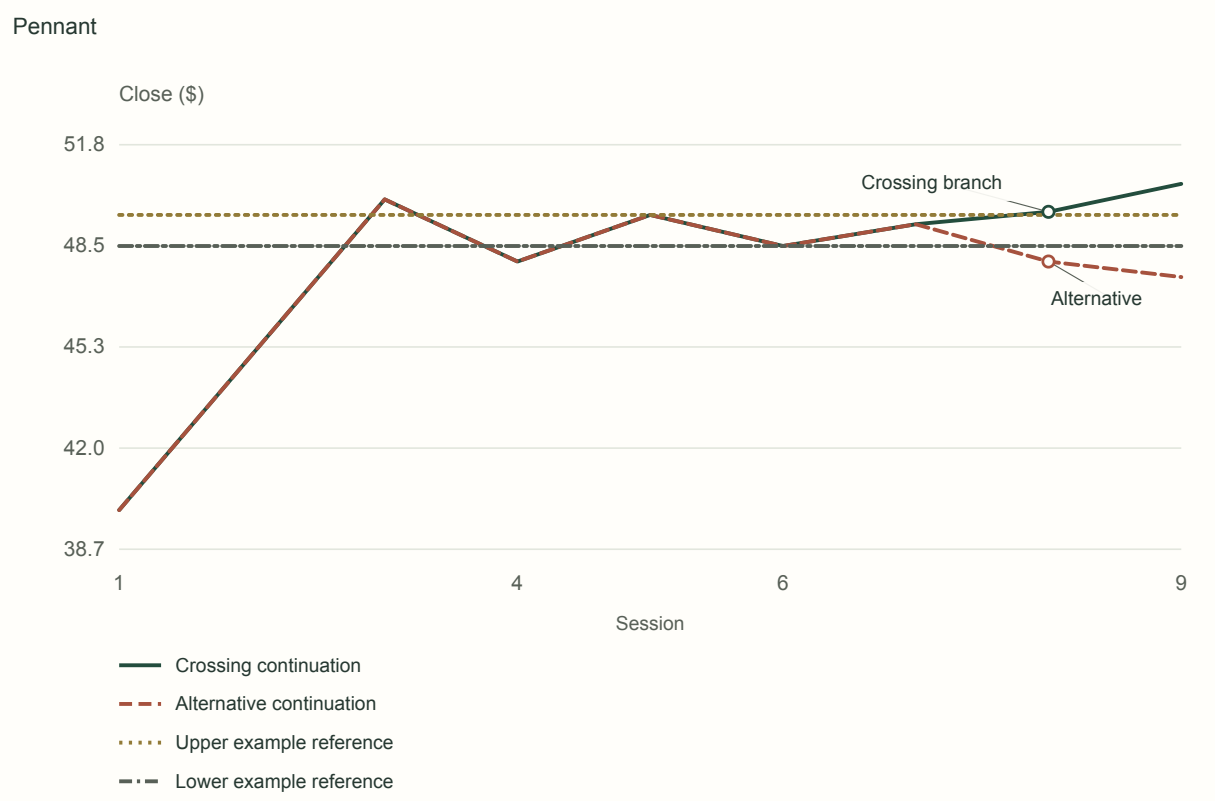


Crossing continuation is downward here; upper 46/lower 44 are final selected references.

Flag



Advance then a short broad parallel pause; the example crossing is above final consolidation high 49.5.



Advance then converging selected swings; boundaries are declared final example references.

Optional-looking catalogue used for the core range/contraction comparison. Each panel holds its prefix fixed and supplies two later branches. Horizontal references are deliberately declared teaching simplifications, not silently projected sloping detectors.

▼ View the illustration's data

Rectangle

Session	Crossing continuation	Alternative continuation	Upper example reference	Lower example reference
1	40	40	44	40
2	44	44	44	40
3	40.2	40.2	44	40
4	43.8	43.8	44	40
5	40.1	40.1	44	40
6	44	44	44	40
7	44.6	39.5	44	40
8	45	39	44	40

Symmetrical triangle

Session	Crossing continuation	Alternative continuation	Upper example reference	Lower example reference
1	50	50	48	46
2	44	44	48	46
3	49	49	48	46
4	45	45	48	46
5	48	48	48	46
6	46	46	48	46
7	48.5	45.5	48	46
8	49	45	48	46

Ascending triangle

Session	Crossing continuation	Alternative continuation	Upper example reference	Lower example reference
1	50	50	50	48
2	44	44	50	48
3	50	50	50	48
4	46	46	50	48
5	50	50	50	48
6	48	48	50	48
7	50.5	47.5	50	48
8	51	47	50	48

Descending triangle

Session	Crossing continuation	Alternative continuation	Upper example reference	Lower example reference
1	50	50	46	44
2	44	44	46	44
3	48	48	46	44
4	44	44	46	44
5	46	46	46	44
6	44	44	46	44

Session	Crossing continuation	Alternative continuation	Upper example reference	Lower example reference
7	43.5	46.5	46	44
8	43	47	46	44

Flag

Session	Crossing continuation	Alternative continuation	Upper example reference	Lower example reference
1	40	40	49.5	48
2	45	45	49.5	48
3	50	50	49.5	48
4	49	49	49.5	48
5	48	48	49.5	48
6	49.5	49.5	49.5	48
7	48.5	48.5	49.5	48
8	50	47.5	49.5	48
9	51	47	49.5	48

Pennant

Session	Crossing continuation	Alternative continuation	Upper example reference	Lower example reference
1	40	40	49.5	48.5
2	45	45	49.5	48.5
3	50	50	49.5	48.5
4	48	48	49.5	48.5
5	49.5	49.5	49.5	48.5
6	48.5	48.5	49.5	48.5
7	49.2	49.2	49.5	48.5
8	49.6	48	49.5	48.5
9	50.5	47.5	49.5	48.5

Optional reference: pause and reversal families · Optional extension

A flag is a short approximately parallel consolidation after a sharp move; a pennant has converging consolidation boundaries after that move. The catalogue uses a 40→50 advance before each pause. It places the prospective closing boundary at the final consolidation high for the example. A later close above it illustrates completion of that crossing condition; a lower continuation supplies a failed continuation candidate. The length of the pause and the importance of the earlier move remain choices in a practical detector.

The double-bottom panel mirrors the core double-top D1–D11 candidate and completed branch A by subtracting each close from 110. Its first and second troughs are 50 and 50.50, with an intervening 56 peak. A 56.50 close crosses the upward reference. The separately supplied alternative uses a 49 close to undercut the first trough; an exact mirror of core branch B's 61.50 would instead be 48.50, which also undercuts 50. This geometric comparison does not imply that the two directions have the same empirical outcomes.

For head and shoulders, the top catalogue shows peaks 60, 65, 61, and intervening troughs 55 at D4 and 56 at D8. Their neckline rises$(56 - 55)/4 = \$0.25$ per step and reaches 57 at D12; a 54 close is below it. A competing 66 close makes a new high instead. The inverted panel supplies the mirror for a bottom. These are reference drawings, not complete detectors: shoulder tolerance and minimum spacing would need an explicit choice before scanning unfamiliar data.

For a core first reading, recognize the geometry and continue to the integrated breakout chapter. Return to this catalogue when commentary uses a name you need to decode. The part review tests construction and transition rather than requiring a large set of labels.

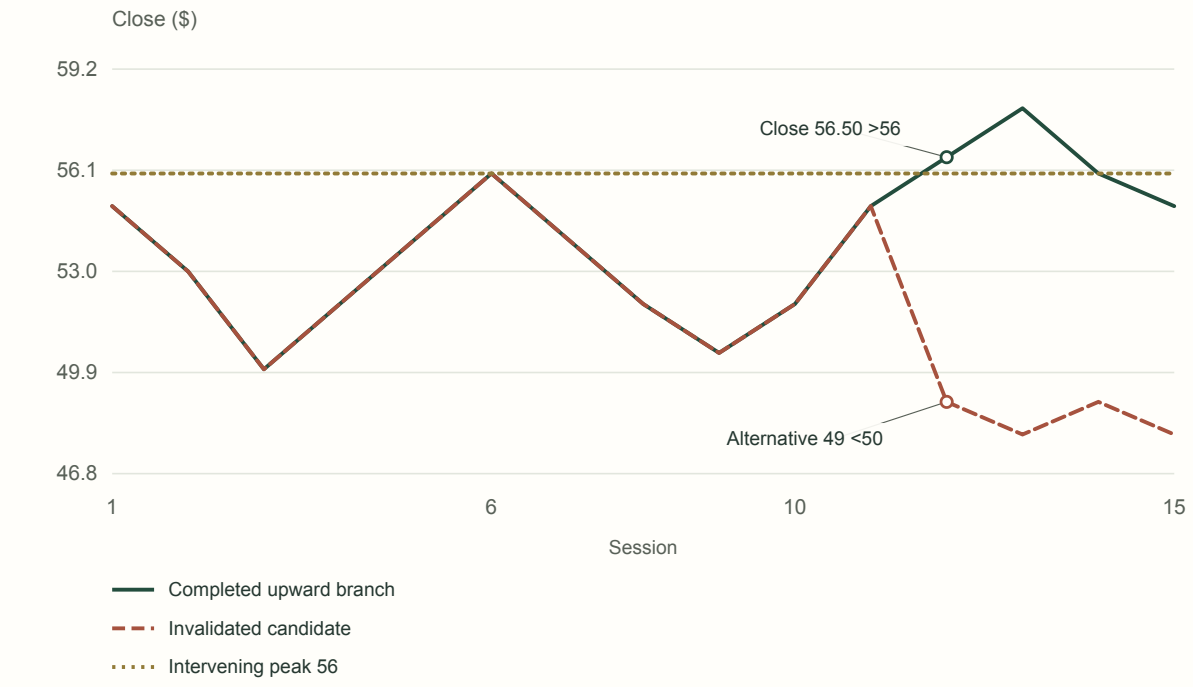
Family reference: what must be specified

Family	Geometric feature	Additional practical choice
Flag	Near-parallel short pause after a move	Move threshold, pause length, boundary crossing
Pennant	Converging short pause after a move	Anchors, degree of convergence, time limit
Double bottom	Two troughs and intervening peak	Trough tolerance, recognition, upward completion
Head/shoulders top	Middle peak above shoulders; trough neckline	Shoulder tolerance, dated neckline, completion
Head/shoulders bottom	Inverted top geometry	Equivalent explicit choices, not assumed symmetry of returns

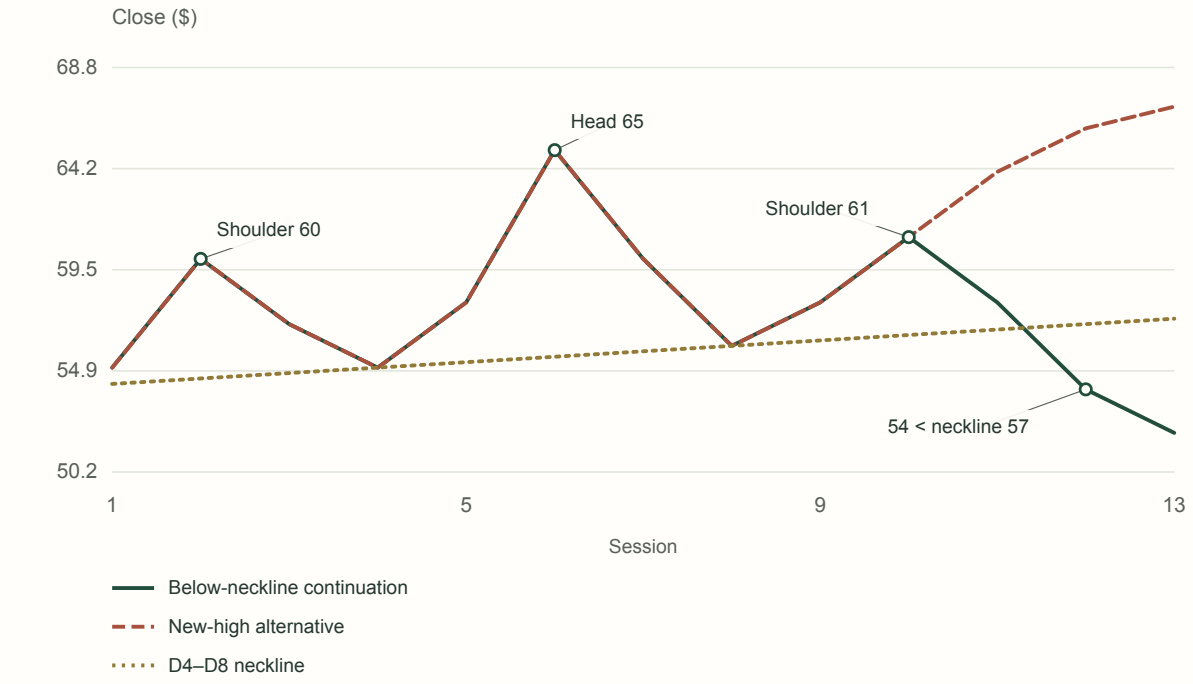
FICTIONAL TEACHING ILLUSTRATION

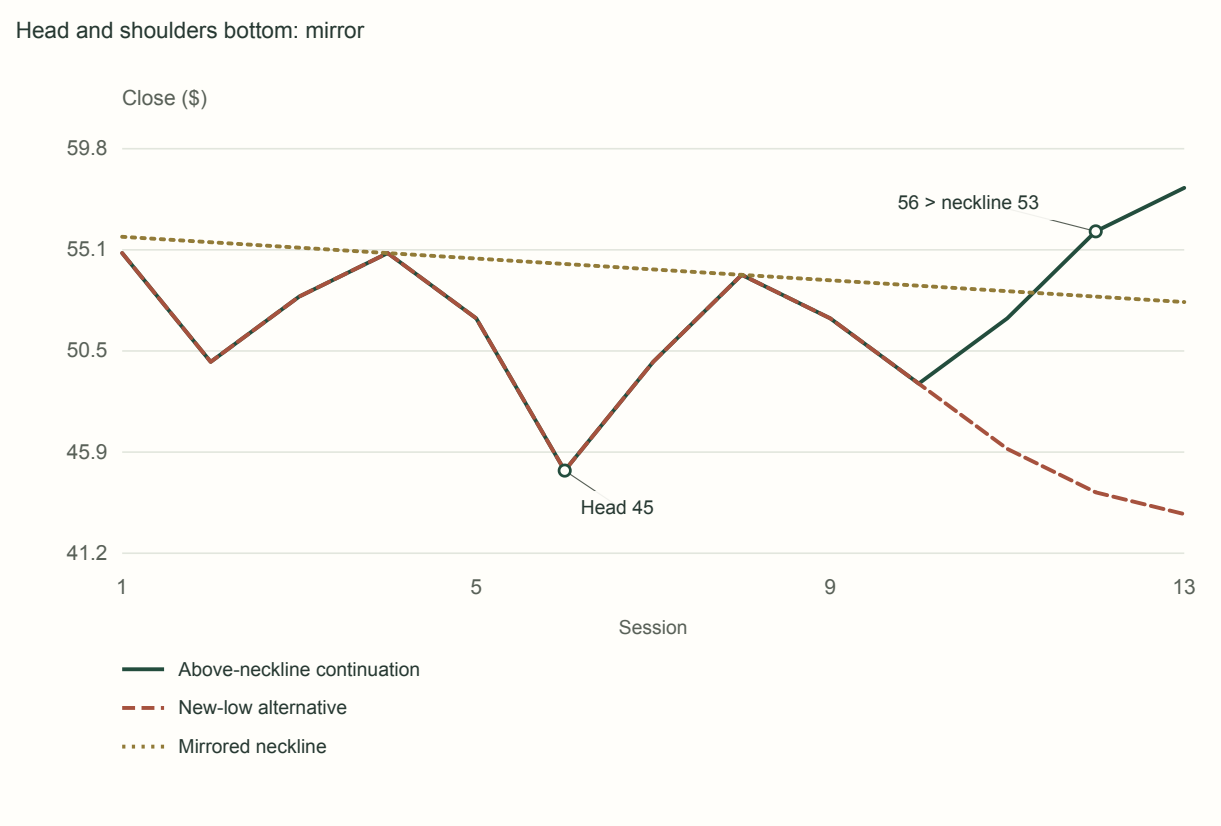
Optional mirrored and multi-peak reference families

Double-bottom mirror



Head and shoulders top





Each case supplies one candidate prefix and two branches. The double-bottom candidate and completed branch mirror the core closes as 110 minus price; its invalidated branch is separately supplied. Head/shoulders diagrams use the dated 55/56 trough neckline (inverted D4 = $55/D8 = $54 for the bottom); these catalogue sketches do not specify a full recognition detector.

▼ View the illustration's data

Double-bottom mirror

Session	Completed upward branch	Invalidated candidate	Intervening peak 56
1	55	55	56
2	53	53	56
3	50	50	56
4	52	52	56
5	54	54	56
6	56	56	56
7	54	54	56
8	52	52	56
9	50.5	50.5	56

Session	Completed upward branch	Invalidated candidate	Intervening peak 56
10	52	52	56
11	55	55	56
12	56.5	49	56
13	58	48	56
14	56	49	56
15	55	48	56

Head and shoulders top

Session	Below-neckline continuation	New-high alternative	D4–D8 neckline
1	55	55	54.25
2	60	60	54.5
3	57	57	54.75
4	55	55	55
5	58	58	55.25
6	65	65	55.5
7	60	60	55.75
8	56	56	56
9	58	58	56.25
10	61	61	56.5
11	58	64	56.75
12	54	66	57
13	52	67	57.25

Head and shoulders bottom: mirror

Session	Above-neckline continuation	New-low alternative	Mirrored neckline
1	55	55	55.75
2	50	50	55.5
3	53	53	55.25
4	55	55	55

Session	Above-neckline continuation	New-low alternative	Mirrored neckline
5	52	52	54.75
6	45	45	54.5
7	50	50	54.25
8	54	54	54
9	52	52	53.75
10	49	49	53.5
11	52	46	53.25
12	56	44	53
13	58	43	52.75

What the drawing has established, and what it has not

The double-top example has enough data to establish construction, timing, completion and one specific subsequent closing outcome. It has no sample of comparable opportunities from which to estimate a useful average trade. That is a concrete distinction between recognizing a pattern and evaluating a strategy. The relevant empirical question would name a detector, market sample, comparator, outcome horizon and costs, rather than ask whether the word double top is generally reliable.

Lo, Mamaysky and Wang used algorithmic recognition in historical US stock data and reported incremental information in some pattern-conditioned return distributions. Their result motivates reproducible questions; it does not assign a success probability to our hand-built pattern or validate this three-session target condition. The book's evidence chapters discuss broader research and how to compare results. In this chapter, the tables and frozen branches provide the smaller prerequisite: a classification that another reader can reproduce and that can actually fail.

A WORKED EXAMPLE

A completed structure without the specified target outcome

After D11 the 60/59.50 peaks and 54 trough make a recognized candidate. D12 close 53.50 completes its closing break. Height 6 projects the conventional 48 reference; the next three closes 52, 54, 55 do not reach it. In the alternate branch, D12 close 61.50 invalidates the candidate before completion. Each conclusion follows from the supplied prices and a different explicit question.

Pause & try it

Specify a double-top rule with a peak tolerance, minimum separation, intervening trough definition, and completion price field. Explain how future bars used to confirm peaks affect the earliest possible recognition time.

One complete answer is the supplied strict two-neighbor close-based detector, at least four session steps between peaks, tolerance≤1% of first peak, minimum intervening close as reference, and later completed close below it.

For 60 and 59.50, tolerance check is 0.50/60 × 100 = 0.833%; separation D9−D3 = 6. The 54 trough is recognized after D8, and the second peak only after D11. D11 close 55 leaves it a candidate; D12 close 53.50 completes it.

Other tolerances or wick-based detectors can be defensible, but state their recognition delays and apply them before seeing the branch. Future bars confirming D9 cannot make its confirmation available on D9.

GUIDED PRACTICE

Guided: complete the frozen candidate note

Stop the core data after D11. Complete the partly filled note: first peak D3 = $60; second peak D9 = ____; separation = 6 sessions; difference = ____%; intervening reference = ____; current state = ____. Then reveal each D12 branch separately.

Peaks 60 and 59.50; separation 6; difference 0.833%; intervening minimum 54. Both peaks and the trough are recognized by D11, but close 55 is not below 54, so the pattern is a candidate.

Branch A53.50 completes the break. Branch B61.50 exceeds first peak 60 and invalidates the candidate. The pre-reveal state must remain identical for both.

INDEPENDENT PRACTICE

Independent: a borderline pair

Two confirmed peaks are 72 and 71.20, separated by five steps, with trough 66. Use the core 1% tolerance and four-step minimum. Does this pair qualify? What changes under a 1.5% alternative? What would a later 65.50 close establish under each detector?

Difference $0.80/72 \times 100 \approx 1.111\%$, exceeding 1%. Separation passes, but the core detector rejects the pair. Under 1.5%, it becomes an eligible candidate.

A65.50 close is below 66. Under the alternative detector it completes the candidate; under the original detector there was no qualifying candidate event to complete. Choose the tolerance before seeing the break rather than relaxing it to capture the favorable case.

CHECK YOUR UNDERSTANDING

One question before you go

D9's second peak is confirmed after D11; D11 close is 55 and D12 close 53.50 versus the 54 reference. When is the core pattern complete?

A. D12, when the required closing break occurs

B. D9, when the second high first appears

C. D11, when both peaks are confirmed

Answer: A. Recognizing both peaks creates a candidate; the later boundary transition completes this detector's event.

Sources & further reading

- Fidelity · Identifying chart patterns with technical analysis ↗
  Engulfing-body vocabulary, triangle categories, breakout filters, false breaks, and retracements. This book does not adopt the slides' performance claims. Original access recorded 2026-10-01. Latest access check 2026-10-12: retrieved.
- StockCharts ChartSchool · Double top reversal ↗
  Two-peak construction, intervening support break, and conventional measured-move calculation. Original access recorded 2026-10-01. Latest access check 2026-10-12: retrieved.
- StockCharts ChartSchool · Head and shoulders top ↗
  Shoulder/head/neckline construction and the need to distinguish formation from completion. Original access recorded 2026-10-01. Latest access check 2026-10-12: retrieved.
- StockCharts ChartSchool · Flag and pennant ↗
  Short consolidation after a sharp move; flag versus pennant geometry. Original access recorded 2026-10-01. Latest access check 2026-10-12: retrieved.
- StockCharts ChartSchool · Symmetrical triangle ↗
  Converging lower highs and higher lows; distinction between pattern geometry and observed breakout direction; filter and retest conventions. No performance percentages are adopted here. Original access recorded 2026-10-01. Latest access check 2026-10-12: retrieved.
- Lo, Mamaysky & Wang · Foundations of technical analysis (NBER working paper 7613) ↗
  Research on formalized pattern recognition and incremental information in a historical US-stock sample. Does not establish a universal, current, after-cost trading edge. Original access recorded 2026-10-01. Latest access check 2026-10-12: retrieved.
- Lo, Mamaysky & Wang (2000), Foundations of Technical Analysis ↗
  Algorithmic pattern recognition and conditional return distributions, distinct from net trading profitability. Original access recorded 2026-10-01. Latest access check 2026-10-12: retrieved.

Teaching examples are fictional unless labelled historical. Inspect this chapter's evidence. Sources support the specified concepts; they do not validate a profitable trading strategy.

Trends, swings & changing structure ↗ Support, resistance & price areas ↗

Breakouts, pullbacks & failed moves ↗ Evidence before conviction ↗

Practice & review this part →